FH | JOANNEUM
eHealth

Artificial Intelligence in Health Care

Exercise 2

EHT25 WS 2026/27

Convolutional Neural Networks for Image Classification

General Information

The task is to implement an image classifier for the classification of colorectal cancer (CRC) based on histological slides. The dataset consists of **89,996** 28×28 colour images for training, **10,004** validation images and **5,744** test images. There are **9 classes** of tissue type: ADI (adipose tissue), BACK (background), DEB (debris), LYM (lymphocytes), MUC (mucus), MUS (smooth muscle), NORM (normal colon mucosa), STR (cancer-associated stroma), TUM (colorectal adenocarcinoma epithelium).

Dataset. The prepared data file is provided on Moodle. Download `path.npz` from there.

Load it with `numpy.load` and extract the splits by key. You can list the available keys with `list(data.keys())`:

```
import numpy as np

data = np.load('path.npz')
print(list(data.keys()))

x_train, y_train = data['train_images'], data['train_labels']
x_val,  y_val  = data['val_images'], data['val_labels']
x_test, y_test = data['test_images'], data['test_labels']
```

1 Data Pre-Processing and Partitioning [2P]:

Make yourself familiar with the dataset by plotting one example image per class, arranged in a figure of e.g. two rows, including class names and labels. Prepare class labels for the use with a neural network and multi-class classification. The images are given in a range of $[0, 255]$; normalise them to $[0, 1]$. Report the class distribution of the training set. You will need it when you interpret the confusion matrix in tasks 2 and 3.

2 Fully-Connected Neural Network [3P]:

First implement a dense neural network with 3 hidden layers (1024, 512 and 128 units) with `relu` activation and an output layer using `softmax` activation. For training, use the categorical cross-entropy loss, early stopping on the validation set $\mathcal{V}$, and e.g. the *Adam* optimizer with an initial learning rate of $\alpha = 0.001$. Consider the use of `Dropout` layers.

In contrast to Exercise 1, the input data here are images. To pass them to a dense network, flatten them in the input layer using `keras.layers.Flatten`.

Plot the learning curves for the training and validation set (loss and accuracy). Evaluate the classification performance (accuracy, confusion matrix, AUROC) on the test set. **Report the number of trainable parameters** to compare it against the CNN.

3 Convolutional Neural Network [5P]:

Now implement a Convolutional Neural Network for classification of the 9 tissue types. In the first stage use a convolutional architecture built from `Conv2D` and `MaxPooling2D` layers. In the second stage use a fully-connected classification network. Do not forget to flatten the output of the convolutional stage before passing it to the fully-connected part. Again use the categorical cross-entropy loss and early stopping on $\mathcal{V}$.

Plot the learning curves for the training and validation set. Evaluate the classification performance (accuracy, confusion matrix, AUROC) on the test set. Try to achieve an accuracy as high as possible (**at least 70 %**).

Summarise the performance measures from tasks 2 and 3 in a table and interpret the results. **Include the parameter count for both models** and comment on what you find.

4 Challenge [2P* Bonus]:

Try to further improve the performance of the CNN by investigating e.g. advanced techniques such as *transfer learning*, weight decay, batch normalization and/or *data augmentation*. Everything is permitted, provided you explain why it was used.

- **[+0.5P]** if accuracy on the test set is above 75 %
- **[+0.5P]** if accuracy on the test set is above 80 %

And the winner takes it all: the group with the highest-scoring model (accuracy and AUROC) wins this bonus task and earns another point **[+1P]**. Save the final model as `cnn_gr_name1_name2.keras` using `your_model.save("cnn_gr_name1_name2.keras")` and submit it together with your report. The model will be evaluated on an external test set.

In addition, each member of the winning group earns the prestigious title:

"Master of Convolutional Neural Networks"